Model Architecture: Qwen2.5-0.5B-Instruct

Generated by sg_modelload

July 21, 2026

1 Model Overview

Model Name	Qwen2.5-0.5B-Instruct
Path	/models/sayak_da_work/Qwen2.5-0.5B-Instruct
Total Parameters	494,032,768
Weight Dtype	bfloat16

2 Forward Pass Algorithm

Algorithm 1 Forward pass of Qwen2.5-0.5B-Instruct

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 896}$
for $layer \leftarrow 0$ **to** 23 **do**
  $h \leftarrow \text{layer}_{layer}(h)$
end for
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 896}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 151936}$

3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.0	Qwen2DecoderLayer	–	–	–
model.layers.0.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.0.mlp	Qwen2MLP	–	–	–
model.layers.0.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.0.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.0.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.0.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.0.self_attn	Qwen2Attention	–	–	–
model.layers.0.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.0.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.0.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.0.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.0.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.0.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.0.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.1	Qwen2DecoderLayer	–	–	–
model.layers.1.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.1.mlp	Qwen2MLP	–	–	–
model.layers.1.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.1.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.1.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.1.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.1.self_attn	Qwen2Attention	–	–	–

Continued on next page

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.1.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.1.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.1.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.1.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.1.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.1.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.1.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.2	Qwen2DecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.2.mlp	Qwen2MLP	–	–	–
model.layers.2.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.2.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.2.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.2.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.2.self_attn	Qwen2Attention	–	–	–
model.layers.2.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.2.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.2.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.2.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.2.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.2.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.2.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.3	Qwen2DecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.3.mlp	Qwen2MLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.3.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.3.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.3.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.3.self_attn	Qwen2Attention	–	–	–
model.layers.3.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.3.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.3.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.3.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.3.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.3.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.3.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.4	Qwen2DecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.4.mlp	Qwen2MLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.4.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.4.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.4.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.4.self_attn	Qwen2Attention	–	–	–
model.layers.4.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.4.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.4.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.4.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.4.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.4.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.4.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.5	Qwen2DecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.5.mlp	Qwen2MLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.5.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.5.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144

Continued on next page

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.5.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.5.self_attn	Qwen2Attention	–	–	–
model.layers.5.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.5.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.5.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.5.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.5.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.5.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.5.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.6	Qwen2DecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.6.mlp	Qwen2MLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.6.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.6.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.6.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.6.self_attn	Qwen2Attention	–	–	–
model.layers.6.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.6.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.6.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.6.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.6.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.6.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.6.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.7	Qwen2DecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.7.mlp	Qwen2MLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.7.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.7.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.7.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.7.self_attn	Qwen2Attention	–	–	–
model.layers.7.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.7.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.7.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.7.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.7.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.7.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.7.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.8	Qwen2DecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.8.mlp	Qwen2MLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.8.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.8.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.8.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.8.self_attn	Qwen2Attention	–	–	–
model.layers.8.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.8.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.8.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.8.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.8.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.8.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.8.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.9	Qwen2DecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.9.mlp	Qwen2MLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144

Continued on next page

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.9.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.9.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.9.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.9.self_attn	Qwen2Attention	–	–	–
model.layers.9.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.9.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.9.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.9.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.9.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.9.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.9.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.10	Qwen2DecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.10.mlp	Qwen2MLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.10.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.10.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.10.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.10.self_attn	Qwen2Attention	–	–	–
model.layers.10.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.10.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.10.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.10.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.10.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.10.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.10.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.11	Qwen2DecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.11.mlp	Qwen2MLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.11.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.11.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.11.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.11.self_attn	Qwen2Attention	–	–	–
model.layers.11.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.11.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.11.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.11.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.11.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.11.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.11.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.12	Qwen2DecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.12.mlp	Qwen2MLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.12.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.12.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.12.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.12.self_attn	Qwen2Attention	–	–	–
model.layers.12.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.12.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.12.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.12.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.12.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.12.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.12.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.13	Qwen2DecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896

Continued on next page

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.13.mlp	Qwen2MLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.13.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.13.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.13.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.13.self_attn	Qwen2Attention	–	–	–
model.layers.13.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.13.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.13.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.13.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.13.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.13.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.13.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.14	Qwen2DecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.14.mlp	Qwen2MLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.14.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.14.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.14.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.14.self_attn	Qwen2Attention	–	–	–
model.layers.14.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.14.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.14.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.14.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.14.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.14.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.14.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.15	Qwen2DecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.15.mlp	Qwen2MLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.15.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.15.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.15.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.15.self_attn	Qwen2Attention	–	–	–
model.layers.15.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.15.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.15.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.15.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.15.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.15.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.15.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.16	Qwen2DecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.16.mlp	Qwen2MLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.16.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.16.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.16.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.16.self_attn	Qwen2Attention	–	–	–
model.layers.16.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.16.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.16.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.16.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.16.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.16.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.16.self_attn.v_proj.bias	Linear	128	bfloat16	128

Continued on next page

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.17	Qwen2DecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.17.mlp	Qwen2MLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.17.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.17.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.17.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.17.self_attn	Qwen2Attention	–	–	–
model.layers.17.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.17.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.17.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.17.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.17.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.17.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.17.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.18	Qwen2DecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.18.mlp	Qwen2MLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.18.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.18.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.18.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.18.self_attn	Qwen2Attention	–	–	–
model.layers.18.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.18.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.18.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.18.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.18.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.18.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.18.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.19	Qwen2DecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.19.mlp	Qwen2MLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.19.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.19.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.19.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.19.self_attn	Qwen2Attention	–	–	–
model.layers.19.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.19.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.19.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.19.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.19.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.19.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.19.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.20	Qwen2DecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.20.mlp	Qwen2MLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.20.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.20.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.20.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.20.self_attn	Qwen2Attention	–	–	–
model.layers.20.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.20.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.20.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.20.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.20.self_attn.q_proj.bias	Linear	896	bfloat16	896

Continued on next page

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.20.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.20.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.21	Qwen2DecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.21.mlp	Qwen2MLP	–	–	–
model.layers.21.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.21.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.21.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.21.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.21.self_attn	Qwen2Attention	–	–	–
model.layers.21.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.21.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.21.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.21.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.21.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.21.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.21.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.22	Qwen2DecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.22.mlp	Qwen2MLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.22.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.22.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.22.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.22.self_attn	Qwen2Attention	–	–	–
model.layers.22.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.22.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.22.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.22.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.22.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.22.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.22.self_attn.v_proj.bias	Linear	128	bfloat16	128
model.layers.23	Qwen2DecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.23.mlp	Qwen2MLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	896 × 4864	bfloat16	4,358,144
model.layers.23.mlp.gate_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.23.mlp.up_proj.weight	Linear	4864 × 896	bfloat16	4,358,144
model.layers.23.post_attention_layernorm.weight	Qwen2RMSNorm	896	bfloat16	896
model.layers.23.self_attn	Qwen2Attention	–	–	–
model.layers.23.self_attn.k_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.23.self_attn.k_proj.bias	Linear	128	bfloat16	128
model.layers.23.self_attn.o_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.23.self_attn.q_proj.weight	Linear	896 × 896	bfloat16	802,816
model.layers.23.self_attn.q_proj.bias	Linear	896	bfloat16	896
model.layers.23.self_attn.v_proj.weight	Linear	128 × 896	bfloat16	114,688
model.layers.23.self_attn.v_proj.bias	Linear	128	bfloat16	128
lm_head.weight	Linear	151936 × 896	bfloat16	136,134,656
model	Qwen2Model	–	–	–
model.embed_tokens.weight	Embedding	151936 × 896	bfloat16	136,134,656
model.layers	ModuleList	–	–	–
model.norm.weight	Qwen2RMSNorm	896	bfloat16	896

4 Parameter Count Estimation

Component	Parameters	% of Total
MLP	313,786,368	63.5%
Embedding	136,134,656	27.6%
LM Head	136,134,656	27.6%
Attention	44,067,840	8.9%
LayerNorm	43,904	0.0%
Total	**494,032,768**	**100.0%**

4.1 Per-Layer Parameter Count

Layer	Params	Running Total
model.layers.0.input_layernorm	896	896
model.layers.0.mlp.down_proj	4,358,144	4,359,040
model.layers.0.mlp.gate_proj	4,358,144	8,717,184
model.layers.0.mlp.up_proj	4,358,144	13,075,328
model.layers.0.post_attention_layernorm	896	13,076,224
model.layers.0.self_attn.k_proj	114,816	13,191,040
model.layers.0.self_attn.o_proj	802,816	13,993,856
model.layers.0.self_attn.q_proj	803,712	14,797,568
model.layers.0.self_attn.v_proj	114,816	14,912,384
model.layers.1.input_layernorm	896	14,913,280
model.layers.1.mlp.down_proj	4,358,144	19,271,424
model.layers.1.mlp.gate_proj	4,358,144	23,629,568
model.layers.1.mlp.up_proj	4,358,144	27,987,712
model.layers.1.post_attention_layernorm	896	27,988,608
model.layers.1.self_attn.k_proj	114,816	28,103,424
model.layers.1.self_attn.o_proj	802,816	28,906,240
model.layers.1.self_attn.q_proj	803,712	29,709,952
model.layers.1.self_attn.v_proj	114,816	29,824,768
model.layers.2.input_layernorm	896	29,825,664
model.layers.2.mlp.down_proj	4,358,144	34,183,808
model.layers.2.mlp.gate_proj	4,358,144	38,541,952
model.layers.2.mlp.up_proj	4,358,144	42,900,096
model.layers.2.post_attention_layernorm	896	42,900,992
model.layers.2.self_attn.k_proj	114,816	43,015,808
model.layers.2.self_attn.o_proj	802,816	43,818,624
model.layers.2.self_attn.q_proj	803,712	44,622,336
model.layers.2.self_attn.v_proj	114,816	44,737,152
model.layers.3.input_layernorm	896	44,738,048
model.layers.3.mlp.down_proj	4,358,144	49,096,192
model.layers.3.mlp.gate_proj	4,358,144	53,454,336
model.layers.3.mlp.up_proj	4,358,144	57,812,480
model.layers.3.post_attention_layernorm	896	57,813,376
model.layers.3.self_attn.k_proj	114,816	57,928,192
model.layers.3.self_attn.o_proj	802,816	58,731,008
model.layers.3.self_attn.q_proj	803,712	59,534,720
model.layers.3.self_attn.v_proj	114,816	59,649,536
model.layers.4.input_layernorm	896	59,650,432
model.layers.4.mlp.down_proj	4,358,144	64,008,576
model.layers.4.mlp.gate_proj	4,358,144	68,366,720
model.layers.4.mlp.up_proj	4,358,144	72,724,864
model.layers.4.post_attention_layernorm	896	72,725,760
model.layers.4.self_attn.k_proj	114,816	72,840,576
model.layers.4.self_attn.o_proj	802,816	73,643,392
model.layers.4.self_attn.q_proj	803,712	74,447,104
model.layers.4.self_attn.v_proj	114,816	74,561,920
model.layers.5.input_layernorm	896	74,562,816
model.layers.5.mlp.down_proj	4,358,144	78,920,960
model.layers.5.mlp.gate_proj	4,358,144	83,279,104
model.layers.5.mlp.up_proj	4,358,144	87,637,248
model.layers.5.post_attention_layernorm	896	87,638,144
model.layers.5.self_attn.k_proj	114,816	87,752,960
model.layers.5.self_attn.o_proj	802,816	88,555,776
model.layers.5.self_attn.q_proj	803,712	89,359,488
model.layers.5.self_attn.v_proj	114,816	89,474,304
model.layers.6.input_layernorm	896	89,475,200
model.layers.6.mlp.down_proj	4,358,144	93,833,344
model.layers.6.mlp.gate_proj	4,358,144	98,191,488
model.layers.6.mlp.up_proj	4,358,144	102,549,632
model.layers.6.post_attention_layernorm	896	102,550,528
model.layers.6.self_attn.k_proj	114,816	102,665,344
model.layers.6.self_attn.o_proj	802,816	103,468,160
model.layers.6.self_attn.q_proj	803,712	104,271,872